

NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA

■ Database Engineering CS2008 ■ End Sem Spring 2022-23 ■ Pg 1 ■ FM 50 ■ 3 Hrs

■ Answer ALL questions. ■ All parts of a question MUST be answered together. ■ Mere answers without proper explanation will not fetch marks. ■ Variation in quality of answer will vary the secured marks.

1. *a.* What is extraneous attribute? What is a non-redundant cover of an FD set?
b. Define alternate key and foreign key. Can the value of foreign key be NULL?
c. For a schema $R(A, B, C, D)$ bearing FDs $F=\{A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow B\}$, find if the decomposition of R into $R_1(A, B)$, $R_2(B, C)$, and $R_3(B, D)$ is dependency preserving and/or lossless?
d. What are the necessary conditions for lossless decomposition of a relation into multiple relations?
e. For a relation with n attributes, what can be the maximum number of functional dependencies in the closure of FD set of the relation? Explain when it happens.

[(1+1.5)+(2+1)+(2.5+2.5)+1.5+3]

2. *a.* What is the difference between DROP, DELETE, and TRUNCATE in SQL?
b. Consider a relation $R(ABCD)$ with FD set $\{A \rightarrow BC, B \rightarrow A, A \rightarrow C, A \rightarrow D, D \rightarrow C, C \rightarrow B\}$. Normalize the relation to the maximum normal form as you can.
c. Find the content of tables G, H, I, J, K, L, M, N from the following tables R and S (Table S is blank):

$G \leftarrow R \div S$ $H \leftarrow R \bowtie_{A=D} S$ $I \leftarrow R \bowtie_{A=D} S$ $J \leftarrow R \ltimes_{A=D} S$
 $K \leftarrow R \triangleright_{A=D} S$ $L \leftarrow R \bowtie_{A=D} S$ $M \leftarrow R \bowtie_{A=D} S$ $N \leftarrow S \div R$

R(A, B, C)			S(D, E, F)		
A	B	C	D	E	F
a ₁	b ₁	c ₁			
a ₂	b ₂	c ₁			
a ₂	b ₁	c ₁			
a ₄	b ₃	c ₃			

- d.* Consider a relationship SUPERVISES(supervisor, subordinate) where both the attributes are foreign key to primary key of EMP(eid, name, salary, dateofjoining) to establish recursive relationship.

Write SQL query:

- (i) to display (in ascending order) the eids of the subordinates whose salary is greater than his supervisor.
(ii) to display (in descending order) the eids of the subordinates whose dateofjoining is later than that of his supervisor.
(iii) to display all possible (unordered) combination of two subordinates under same supervisor.
(iv) to print eids of all employees who neither have subordinates nor have supervisors.
(v) to print eids of all employees who either do not have any subordinate or do not have any supervisor.
(vi) to print all eids who have both supervisor and subordinate roles.
(vii) to print all pairs of peers without duplicates (A peer of an employee is defined as another employee at his level of hierarchy, i.e. subordinate of his supervisor, or a subordinate of subordinate of his supervisor's supervisor etc.).

[2.5 + 3 + (0.5 × 8) + [(1.25 × 6) + 3]]

3. *a.* What is controlled concurrency in transactions? What are different types of conflicts?
b. Given the functional dependency (FD) set $F=\{X \rightarrow Z, ZW \rightarrow X, Z \rightarrow X, Z \rightarrow W\}$. Using membership algorithm, determine if the FD $Z \rightarrow X$ is redundant in F .
c. What is isolation property? Explain with an example.
d. Which is better? cascadeless or cascading recovery of schedules? Justify.
e. For k concurrent transactions $T_1, T_2, \dots, T_k$ where T_i has n_i operations, what is the number of all possible non-serial schedules?
f. Differentiate between recoverable, cascadeless, and strict schedules.
g. Check if the following schedule S on three transactions T_1, T_2 , and T_3 is conflict serializable:
 $S : \{R_1(A), W_2(A), R_3(A), W_1(A), W_3(A)\}$

[(1+1.5)+2.5+(1+1)+1.5+1.5+2+3]